E-Commerce Checkout Funnel & Conversion Optimization

Executive Analytical Report | SQL + Power BI Portfolio Strategy (9.24M Sessions)

1. Executive Summary & Core Metrics

This project analyzes an enterprise-scale e-commerce event dataset comprising 9.24 million customer sessions (42M+ raw events) to diagnose cart abandonment drivers and conversion drop-offs. By building a robust PostgreSQL data model and interactive Power BI executive dashboards, the analysis addresses conversion friction points, providing strategic recommendations projected to increase checkout completion rates by 4.2% and recapture over $1.2M in annual GMV.

TOTAL SESSIONS	CART SESSIONS	ABANDONMENT RATE	PROJECTED RECOVERY
9.24M	**561.89K**	**50.05%**	**$1.2M+**
Dataset Scale	Active Buyers	Primary Bottleneck	Annual GMV Recaptured

2. Customer Funnel & Conversion Architecture

A critical modeling decision involved handling 'Purchase Without Cart' anomalies where users completed transactions without explicit cart logs. Isolating these tracking anomalies established a true baseline cart conversion rate of 49.95%.

Funnel Stage	Session Count	% of Total	Key Business Meaning
Viewed Only	8,344,836	90.27%	Top-of-funnel browsing without intent
Cart Abandoned	281,237	3.04%	High-intent users dropping at checkout
Cart -> Purchase	280,650	3.04%	Successfully converted cart buyers
Purchase Without Cart	337,699	3.65%	Direct buy / fast-track checkout logs

3. Behavioral Drivers of Abandonment

Cart size and browsing behavior reveal clear operational friction points. Single-item carts show elevated drop-off rates, while heavy product exploration strongly correlates with user decision paralysis.

Behavioral Metric	Category Bucket	Cart Sessions	Abandoned	Abandonment %
Cart Size Impact	1 item	385,757	207,223	**53.72% (Highest)**
Cart Size Impact	3 items	36,423	14,270	**39.18% (Optimal)**
Product Views	1 view	92,674	44,052	47.53%

Product Views	11+ views	79,062	48,459	**61.29% (Paralysis)**
Repeat View Friction	6+ Repeat Views	76,720	45,346	**59.11% (Hesitation)**

4. High-Priority Category Opportunities

To avoid chasing high percentage drop-offs with low sales impact, category prioritization combines cart volume, abandonment percentage, and repeat-view friction into a weighted opportunity score.

Category Code	Cart Volume	Abandonment %	Repeat Rate	Priority Level
computers.desktop	2,278	60.71%	38.24%	**Critical Opportunity**
construction.tools.drill	2,354	60.11%	38.49%	**Critical Opportunity**
appliances.kitchen.refrigerators	8,386	58.06%	35.40%	**Critical Opportunity**
electronics.smartphone	327,582	44.89%	20.40%	High Volume - Monitor

5. Strategic Action Roadmap & Experimentation Framework

- **Single-Item Checkout Simplification:** Implement 1-click Express Checkout (Apple/Google Pay) and guest options. Target: Recover $1.2M in single-item abandonments.
- **High-Exploration Comparison Tools:** Deploy inline comparison matrices and review summaries for users with 6+ views. Target: Reduce high-exploration drop-off by 15%.
- **Targeted Category Interventions:** Audit pricing, stock visibility, and shipping fees for Action Area categories (`computers.desktop`, `tools.drill`).
- **Controlled A/B Testing Protocol:** Execute hypothesis-driven experiments: Observation -> Hypothesis -> Intervention -> A/B Test -> KPI Evaluation.